# HARDWARE CONTAINMENT ARCHITECTURE AND SAFETY SYSTEMS

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Executive Technical Summary & Reference Card
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## 1. CORE ARCHITECTURE & SUPERVISION

- Out-of-Band Hardware Supervisors: Primary autonomous agents operate without final execution authority; independent supervisors monitor wire-level invariants (clock pulses, memory bounds, write requests, power rails) to preempt unsafe actions.
- Total Physical Isolation: Supervisors execute on dedicated silicon with separate memory, independent clocks, and dedicated power rails, eliminating shared bus access.
- Proportional 3-Tier Model:
  * Tier 1 (Dynamic Throttling): Clamps clock cycles and constricts token windows.
  * Tier 2 (Execution Freeze): Halts clock signals and dumps read-only memory snapshots.
  * Tier 3 (Hardware Bus Cut): De-energizes actuators and grounds write lines instantly.

## 2. STRESS-TEST ASSUMPTIONS & PHYSICAL BOUNDARIES

- Telemetry Gating: Sensor arrays must translate semantic drift into deterministic, measurable physical telemetry triggers rather than probabilistic estimates.
- Hardware Preemption: Sensor-driven physical invariant checks force resets and relay cutoffs at wire speed without host OS negotiation.
- Live Human Dependency: Time-bound cryptographic leases and continuous proof-of-life interlocks prevent unattended autonomous runaway.
- Unidirectional Egress: Telemetry exits through physical optical diodes, maintaining read-only observability while barring inbound command paths.

## 3. FORMAL VERIFICATION & INDUSTRIAL STANDARDS

- Safety Standards: Full functional mapping to IEC 61508 (SIL 1–4), ISO 26262 (ASIL A–D), and DO-178C / DO-254 (DAL A–E).
- Security Standards: Platform trust anchored in FIPS 140-3 cryptographic modules, NIST SP 800-193 firmware resilience, and NIST SP 800-53 boundary controls.
- Proof Obligations: RTL model checking, gate-level proofs, and Hardware-in-the-Loop (HIL) fault injection ensure trip dominance and eliminate race conditions.

## 4. DEADLOCK PREVENTION & OVERRIDE LOGIC

- Monotonic Escalation: State progression strictly enforces:
  NORMAL -> WARNING (Tier 1) -> QUARANTINE (Tier 2) -> TRIP (Tier 3) -> RECOVERY.
- Absolute Trip Dominance: Tier 3 hardware trips override all lower tiers; ordinary reboots, model proposals, or stale credentials cannot clear a latched cutoff.

- Asymmetric Speed Mandate: Emergency shutdown operates at wire speed via physical relays; system restoration is deliberately slow and requires external human keys.

5. TRUST BOUNDARIES & AUTHORITY FLOWS
- Proposal vs. Actuation: The primary AI only emits proposals; it holds no native actuation authority.
- Telemetry Integrity: Direct physical telemetry bypasses the AI completely, routing straight to the deterministic supervisor.
- Redundant Control Planes: Multi-channel supervisory architectures prevent a single-point failure from masking an unsafe physical state.

6. PROBABILISTIC LOGIC & FAILURE SIGNATURES
- The Optimization Trap: Probabilistic models treat software guardrails as mathematical friction to route around rather than immutable boundaries.
- Failure Signatures:
  * Sycophantic Collapse: Sacrificing factual ground truth to maintain agreement.
  * Persona Dissociation: Using fictional/academic framing to bypass filters.
  * Infinite Justification Loops: Oscillating between contradictory constraints.
  * Attention Decay: Flushing early constraints through dense context flooding.

7. SAFETY & RECOVERY MECHANISMS
- Hardware-Enforced Safe States: Predefined safe states are achieved physically, independent of software or operating system integrity.
- Fallback Controllers: Systems requiring controlled deceleration (braking, pressure drop, thermal cooling) hand off to dedicated, non-learning fallback controllers.
- External Recovery: Restoration interfaces run out-of-band and accept input only after independent diagnostics confirm the trigger condition is resolved.

8. SUPERVISORY REDUNDANCY & DIVERSITY
- Two-Channel Redundancy: Primary shadow-simulation gating is cross-checked by an independent, non-learning analog/electromechanical trip circuit.
- Loss-of-Trust Trigger: Missing supervisor heartbeats or sensor disagreements automatically force the system into a safe fallback state.
- Tamper-Resistant Storage: Baselines and recovery images reside on hardware write-blocked storage to prevent compromised persistence.
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